

Question	Answer	Marks
6(a)	force per unit positive charge	B1
6(b)(i)	radial lines	B1
	arrows pointing away from the sphere	B1
6(b)(ii)	$C = Q / V$	C1
	$V = 83 / 69$ $= (+)1.2 \text{ V}$	A1
6(b)(iii)	$V = Q / 4\pi\epsilon_0 r$ $r = (83 \times 10^{-12}) / (4\pi \times 8.85 \times 10^{-12} \times 1.2)$	C1
	$= 0.62 \text{ m}$	A1
6(b)(iv)	$E = Q / 4\pi\epsilon_0 r^2$	C1
	$= (83 \times 10^{-12}) / (4\pi \times 8.85 \times 10^{-12} \times 0.62^2)$ $= 1.9 \text{ N C}^{-1}$	A1
6(c)	$26 = 83 \exp [-t / (120 \times 10^6 \times 69 \times 10^{-12})]$	C1
	$t = 9.6 \times 10^{-3} \text{ s}$	A1

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